# Practice workbook · Identify bacterial cell structures and compare with human cells

Labelling bacterial cells and comparing their parts and functions with human cells such as skin cells or sperm.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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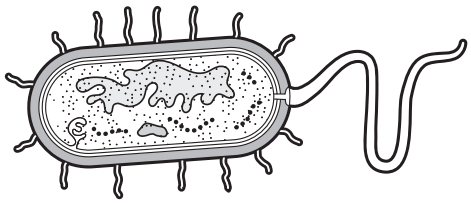
Type ID: HBI-BACTERIAL-CELL-STRUCTURE

# 1. Question 1 · part a.ii · 2 marks

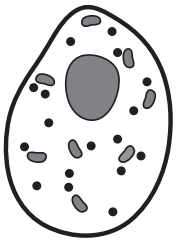
June 2024 Paper 1R

Worked-example retry · Attempt part a.ii only.

**1** (a) The diagrams show a bacterial cell and a human skin cell.



Bacterial cell



Human skin cell

(i) Give **two** similarities between the bacterial cell and the human skin cell.

**(2)**

1 .....

.....

2 .....

.....

(ii) Give **two** differences between the bacterial cell and the human skin cell.

**(2)**

1 .....

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2 .....

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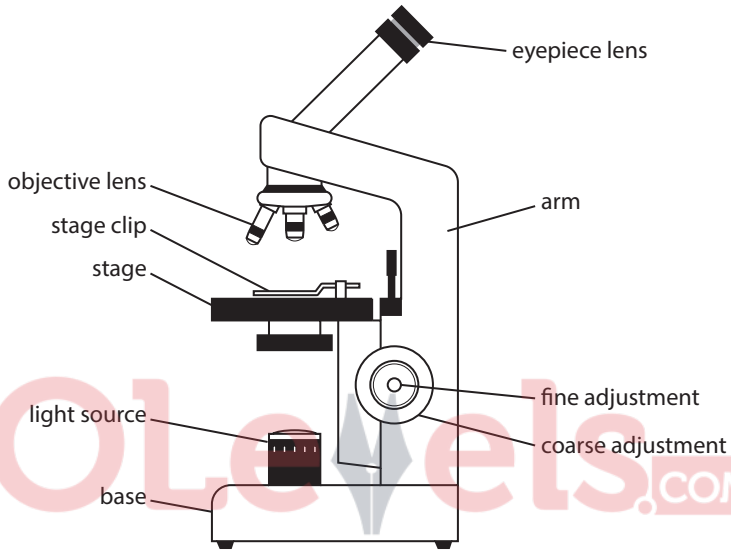
# 1. Question 1 · part a.ii · 2 marks

June 2024 Paper 1R

Worked-example retry · Attempt part a.ii only.

(b) Many structures in cells can be viewed more clearly using a light microscope.

The diagram shows a light microscope.




(i) Draw **one** straight line from each part of the light microscope to its function.

**(3)**

| Part of microscope | Function |
|---|---|
| objective lens | holds the microscope slide in place |
| stage clip | moves the stage to produce a clear, focused image |
| fine adjustment | magnifies the sample |
| | keeps the microscope steady |

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## 1. Question 1 · part a.ii · 2 marks · Working and answer

Working and answer

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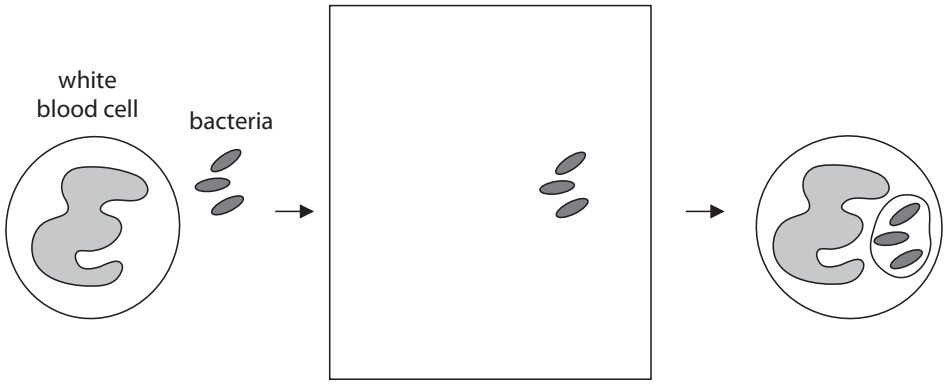
# 2. Question 2 · part b · 3 marks

June 2019 Paper 1

Further practice · Attempt part b only.

**2** (a) Bacteria can cause disease in humans.

The diagram shows how one type of white blood cell helps to defend the body against disease.




(i) Complete the diagram by drawing the shape of the white blood cell at stage 2.

**(1)**

(ii) The box lists words associated with bacteria and disease.

| | | | | | |
|---|---|---|---|---|---|
| acids | enzymes | erythrocytes | lymphocytes | phagocytes | toxins |

Use words from the box to complete the sentences.

**(2)**

White blood cells called ..... engulf bacteria.

These white blood cells contain ..... to digest bacteria.

(iii) One way that white blood cells defend the body from disease is shown in the diagram.

State another way in which white blood cells defend the body against disease.

**(1)**

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# 2. Question 2 · part b · 3 marks

June 2019 Paper 1

Further practice · Attempt part b only.

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(b) The table lists structures found in some cells.

Place ticks in boxes to show which structures are found in bacterial cells and which are found in human skin cells.

One row has been completed for you.

**(3)**

| Structures | Bacterial cell | Human skin cell |
|---|---|---|
| nucleus | | |
| DNA | | |
| cytoplasm | ✓ | ✓ |
| cell wall | | |

(c) Viruses can also cause diseases in humans.

Many viruses contain RNA as their genetic material.

Which statement describes the structure of RNA?

**(1)**

☒ **A** a double-stranded helix containing the bases ATGC

☒ **B** a double-stranded helix containing the bases AUGC

☒ **C** a single-stranded helix containing the bases ATGC

☒ **D** a single-stranded helix containing the bases AUGC

**(Total for Question 2 = 8 marks)**

## 2. Question 2 · part b · 3 marks · Working and answer

Working and answer

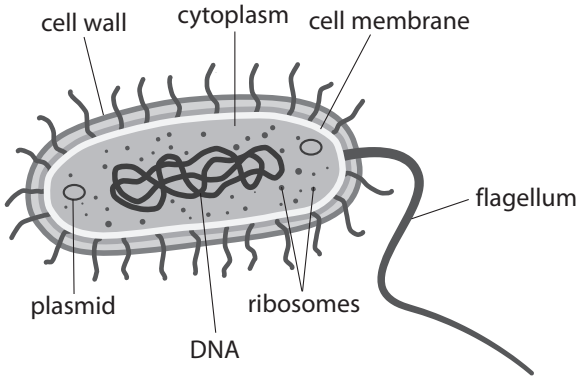
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# 3. Question 3 · part a.i · 3 marks

January 2020 Paper 1

Further practice · Attempt part a.i only.

**3** (a) The diagram shows a type of bacterial cell.




(i) The boxes give some structures found in the bacterial cell and some functions of cell structures.

Draw one straight line from each structure to its correct function.

**(3)**

| Structure | Function |
|---|---|
| cell membrane | protects against pathogens |
| | helps the cell to move |
| ribosomes | makes proteins |
| | holds the genetic code |
| flagellum | |
| | controls which substances enter and exit the cell |

(ii) State two structural differences between the bacterial cell and a sperm cell.

**(2)**

1 .....

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2 .....

.....

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8

P 5 9 9 0 9 A 0 8 2 8

## 3. Question 3 · part a.i · 3 marks · Working and answer

Working and answer

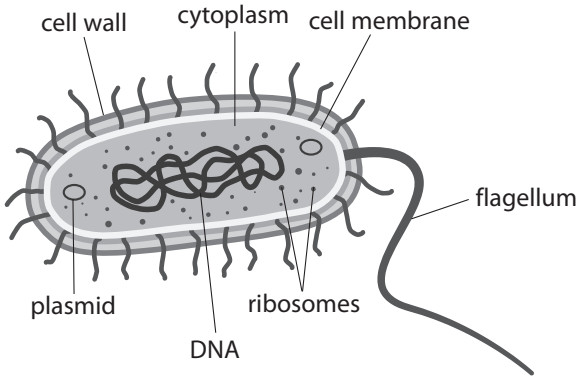
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# 4. Question 3 · part a.ii · 2 marks

January 2020 Paper 1

Further practice · Attempt part a.ii only.

**3** (a) The diagram shows a type of bacterial cell.




(i) The boxes give some structures found in the bacterial cell and some functions of cell structures.

Draw one straight line from each structure to its correct function.

**(3)**

| Structure | Function |
|---|---|
| cell membrane | protects against pathogens |
| | helps the cell to move |
| ribosomes | makes proteins |
| | holds the genetic code |
| flagellum | controls which substances enter and exit the cell |

(ii) State two structural differences between the bacterial cell and a sperm cell.

**(2)**

1 .....

.....

2 .....

.....

**8**

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# 4. Question 3 · part a.ii · 2 marks

January 2020 Paper 1

Further practice · Attempt part a.ii only.

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(b) Many bacteria absorb oxygen and nutrients from their environment. Oxygen and nutrients are used by bacteria in aerobic respiration.

(i) Which process transports oxygen into bacterial cells?

**(1)**

- ☒ **A** active transport
- ☒ **B** diffusion
- ☒ **C** excretion
- ☒ **D** osmosis

(ii) Complete the word equation for aerobic respiration.

**(2)**

oxygen + ..... → ..... + water

(iii) Explain why respiration is important for bacteria.

**(2)**

.....

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.....

.....

**(Total for Question 3 = 10 marks)**

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## 4. Question 3 · part a.ii · 2 marks · Working and answer

Working and answer

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## 5. Question 4 · part a.i · 1 mark

January 2023 Paper 1

Further practice · Attempt part a.i only.

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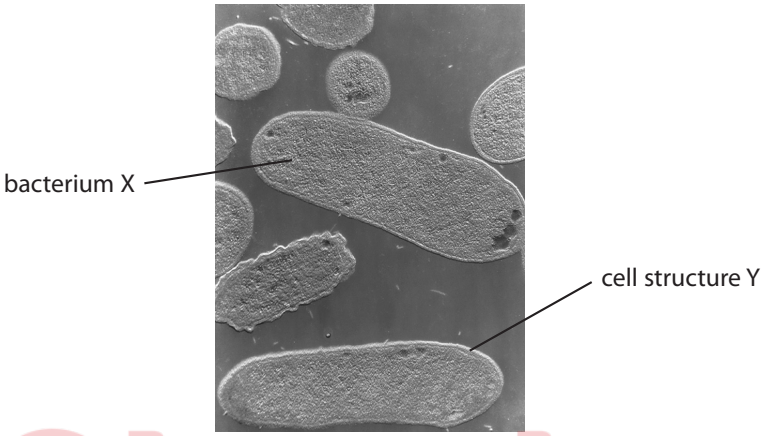
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**4** Typhoid is caused by *Salmonella typhi* bacteria.

These bacteria are spread by houseflies.

(a) The photograph shows *Salmonella typhi*, magnified 30 000 times.




(Source: © CNRI/SCIENCE PHOTO LIBRARY)

(i) Each bacterium is surrounded by cell structure Y.

Identify cell structure Y.

**(1)**

.....

(ii) On the photograph bacterium X has a length of 55 mm.

Calculate the actual length of the bacterium in micrometres (µm).

[1 mm = 1000 µm]

**(2)**

actual length = ..... µm

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## 5. Question 4 · part a.i · 1 mark · Working and answer

Working and answer

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## 6. Question 8 · part a.i · 3 marks

June 2023 Paper 1R

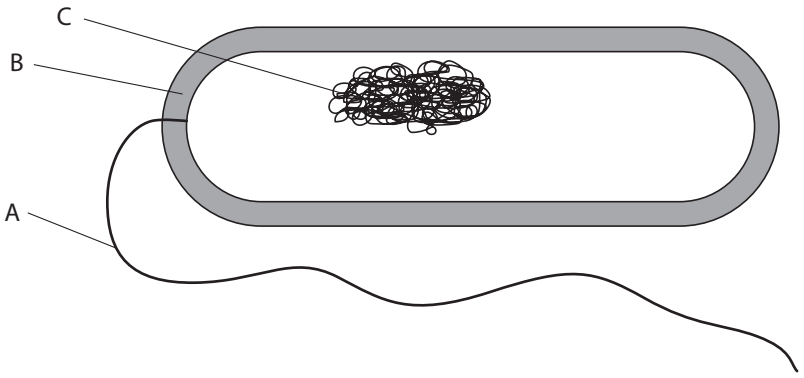
Further practice · Attempt part a.i only.

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**8** The diagram shows a bacterial cell.




(a) (i) Name the parts labelled A, B and C.

**(3)**

A .....

B .....

C .....

(ii) Some types of bacteria cause disease.

Give the name of organisms that cause disease.

**(1)**

.....

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## 6. Question 8 · part a.i · 3 marks · Working and answer

Working and answer

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# 7. Question 1 · part a.ii · 1 mark

June 2023 Paper 2R

Further practice · Attempt part a.ii only.

**1** (a) The table shows some cell structures and their functions.

(i) Place one tick (✓) in each row to show the function of each cell structure.

**(3)**

| Cell structure | Function | | |
|---|---|---|---|
| | **makes proteins** | **contains DNA** | **allows substances in and out of cells** |
| nucleus | | | |
| ribosomes | | | |
| cell membrane | | | |

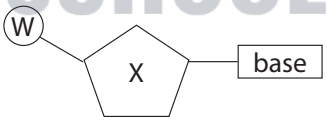
(ii) Give a structure found in bacterial cells.

**(1)**

.....

(b) (i) DNA is made of two chains of nucleotides.

The diagram shows one nucleotide.




Name part W and part X.

**(2)**

W .....

X .....

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## 7. Question 1 · part a.ii · 1 mark · Working and answer

Working and answer

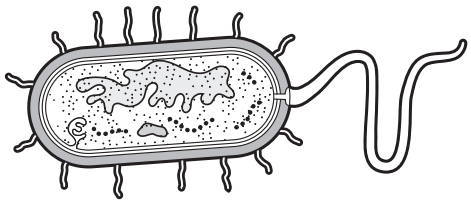
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# 8. Question 1 · part a.i · 2 marks

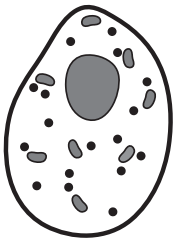
June 2024 Paper 1R

Further practice · Attempt part a.i only.

**1** (a) The diagrams show a bacterial cell and a human skin cell.



Bacterial cell



Human skin cell

(i) Give **two** similarities between the bacterial cell and the human skin cell.

**(2)**

1 .....

.....

2 .....

.....

(ii) Give **two** differences between the bacterial cell and the human skin cell.

**(2)**

1 .....

.....

2 .....

.....

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**2**

## 8. Question 1 · part a.i · 2 marks · Working and answer

Working and answer

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# 9. Question 4 · part b.i · 3 marks

June 2025 Paper 1

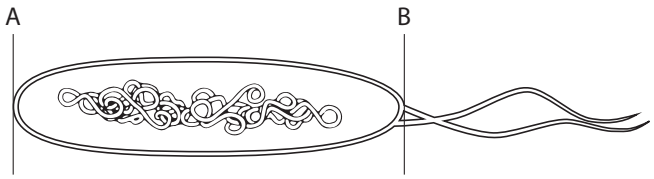
Further practice · Attempt part b.i only.

**4** (a) State what is meant by the term **pathogen**.

**(1)**

.....

(b) The diagram shows a pathogenic bacterium.




(i) Label **three** structures shown on the diagram.

**(3)**

(ii) The actual length of the bacterium between points A and B is 2.0 µm.

Calculate the magnification of the diagram.

[1 mm = 1000 µm]

**(3)**

magnification = × .....

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## 9. Question 4 · part b.i · 3 marks · Working and answer

Working and answer

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